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Transformer Based Sentiment Analysis on Code Mixed Data

Koyyalagunta Krishna Sampath and Supriya M.

Department of Computer Science and Engineering,
Amrita School of Computing, Amrita Vishwa Vidyapeetham, Bengaluru,
Karnataka, 560035, India
bl.en.p2cse21012@bl.students.amrita.edu
m_supriya@blr.amrita.edu

Abstract

In India, a country known for its linguistic diversity, code mixing is a common practice, and it has a profound impact on the way people communicate through various mediums, including social media platforms and everyday conversations. The prevalence of code-mixing in social media platforms presents a substantial hurdle for machine translation and language processing tasks. The abundance of unstructured text in code-mixed form on these platforms highlights a crucial research domain within NLP. The blending of Hindi and English, known as Hinglish, and other mixed case text like Malayalam-English, Tamil-English, Telugu-English are particularly prevalent among the younger generation while communication in social media and requires appropriate processing to aid comprehension by both monolingual users and language processing models. Manual translation of this type of data proves to be laborious due to challenges like limited vocabulary, potential misunderstandings of context, grammatical errors, biases, and various other issues. Additionally, existing translation models tend to perform more effectively on monolingual language rather than code-mixed data. Therefore, it is more desirable to build models that can translate code-mixed data.
This study tries to convert code-mixed Hinglish, Malayalam-English, Tamil-English, Telugu-English language in Romanised script to monolingual English which can further be given as input to NLP applications like Sentiment Analysis. This is achieved by finetuning pretrained models like IndicLID for Language Identification (LID) module and use an ensemble approach for transliteration + translation using Indictrans and IndicXlit for code mixed machine translation which will be given as input to classification algorithm which performs Sentiment Analysis and predict the sentiment. It is observed that this approach of translation of code-mixed test perform better than traditional machine translation for Indian languages Hindi, Tamil, Telugu and Malayalam.

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Keywords: Natural Language Processing; Code Mixing; Language Identification; Sentiment Analysis; Translation; Transliteration,; Transformers

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1. Introduction

Code mixing is a linguistic phenomenon characterized by the blending of multiple languages within a single communication instance. It has received a great deal of interest in the field of Natural Language Processing (NLP) due to its prevalence in multilingual societies, particularly in countries like India. India is a linguistically diverse nation, with a rich tapestry of languages spoken across its regions. While there are 22 officially recognized languages in India, there are several hundred regional languages and dialects in use. There are several reasons behind the prevalence of code mixing in India. One prominent factor is the widespread influence of English due to historical, colonial, and post-independence factors. English has become a language of prestige and aspiration, and its integration with regional languages allows individuals to showcase their proficiency in both English and their native language.

Additionally, code-mixing serves as a form of cultural identity and expression. It allows individuals to seamlessly switch between languages based on the context, social group, or emotional tone they wish to convey. By blending languages, people can create a sense of inclusivity, bridging the gap between different linguistic communities and facilitating communication among diverse groups. There are different types and forms of code mixing found in social media for Indian languages such as insertional code mixing (inserting words from one language into another), alternating code mixing (switching between languages at a sentence or clause level), congruent lexicalization (using lexical items from one language within the grammatical structure of another), calquing (direct translation of idiomatic expressions), code switching (switching between languages within a conversation), and phrasal code mixing (combining words or phrases from different languages to form a new phrase).

Code mixing in Roman/Latin script and Devanagari script are two distinct approaches to representing Indian languages using the English alphabet and native scripts, respectively. While both scripts facilitate code mixing and communication in Indian languages, they pose unique challenges for Natural Language Processing (NLP) systems. Roman/Latin script transliterates Indian words into English characters, making it accessible to users familiar with the English alphabet. However, this transliteration can lead to inconsistencies and variations in spelling, complicating NLP tasks like sentiment analysis and machine translation. Devanagari script, on the other hand, represents Indian words directly in their native script. While this approach preserves the original language structure, it can be challenging for NLP systems that are primarily trained on Roman/Latin script data. Both Roman/Latin script and Devanagari script code mixing introduce complexities for NLP, requiring specialized techniques to effectively process and analyse code-mixed data. These techniques need to address the linguistic variations, grammar differences, and inconsistent language structures inherent in code-mixed text. The presence of multiple languages within a single text poses specific challenges for sentiment analysis, as sentiment polarity and vocabulary usage can differ significantly across languages. Additionally, the limited availability of code-mixed data and the inherent complexity of linguistic factors further hinder accurate sentiment classification.

To effectively handle code-mixed data, NLP systems require advancements in language modelling, cross-lingual representation learning, and adaptation to multilingual contexts. Overcoming these challenges will be crucial for developing NLP applications that can accurately process and analyse code-mixed text, enabling improved communication and information access for multilingual users. Despite the challenges posed by code-mixing in NLP, it also presents exciting opportunities for innovation and research. Developing models and techniques that can effectively handle code-mixed data has become a crucial area of focus in NLP, particularly in the context of India and other multilingual societies. Advancements in this field can lead to more accurate sentiment analysis, improved machine translation between languages, and a deeper understanding of the complexities of code-mixed communication. Transformers have emerged as a powerful and versatile neural network architecture for various NLP tasks, including language translation, text classification, and language modelling. Unlike traditional neural networks, transformers process input data as sequences of tokens, such as words or sub-words, and produce either a sequence of tokens or a fixed-length representation of the input. This unique approach enables transformers to efficiently handle long data sequences and capture relationships between tokens that may be distant in the sequence.

The training of transformers typically involves using large amounts of labelled data and a variant of stochastic gradient descent. These advanced training techniques have led to remarkable performance levels, often considered "state-of-the-art," in various NLP tasks. Consequently, transformers have gained popularity as a preferred choice for NLP applications due to their effectiveness and efficiency. One of the key strengths of transformers is their ability to

extract context from sentences rather than relying solely on keywords. This capability distinguishes them from conventional language processing approaches and makes them well-suited for handling the nuances of code-mixed data.

This work acknowledges the work already done and tries to carry forward the research done by addressing the limitations observed in previous works for code-mixed translation for Indian languages. This study tries to convert Romanized code-mixed Hindi-English, Malayalam-English, Tamil-English, and Telugu-English language to English which can further be given as input to NLP applications like Sentiment Analysis. This is achieved by finetuning pre-trained models like Indic LID for Language Identification (LID) module and using an ensemble approach for transliteration + translation using Indic trans and IndicXlit for code mixed machine translation which is given as input to classification algorithm which performs Sentiment Analysis and predict the sentiment. The above four languages constitute approximately 60% of code-mixed languages observed on social media. This study compares this proposed approach with existing machine translation tool like Google Translate and deduces that a transformer model specifically trained for code mixed translation perform much better than traditional machine translation for Indian languages Hindi, Tamil, Telugu, and Malayalam. This can further be extended beyond sentiment analysis and used in other applications like customer service chatbots to help businesses communicate with customers who use code-mixed language. It could also be used in educational settings to help students who are not fluent in English understand lectures and course materials.

2. Literature review

There were a lot of studies done proposing several solutions and frameworks for working with code mixed data in Indian Languages mostly dominated by Hinglish.

The paper [1] addresses the translation challenge of Hinglish conversations to English, focusing on social media communication. The "Code-Mixed Hinglish to English Language Translation Framework" incorporates Language Identification, Segmentation, and Translation components to improve translation accuracy. It outperforms Google's Neural Machine Translation alone, demonstrating its effectiveness in translating Hinglish. The work in [2] proposes a deep learning approach for sentiment analysis of Hindi-English code-mixed text. Their method leverages both character and word features through a dual-component architecture, utilizing pre-trained word embeddings. The approach achieves impressive results on Hinglish and Code-Mixed Sentiment Analysis Datasets. Study in [3] focuses on sentiment analysis of code-mixed Hindi-English social media text. The authors propose a bidirectional LSTM-based architecture and utilize ensemble techniques to enhance their model's accuracy. Their system achieves better performance than current state-of-the-art systems. The paper [4] presents a deep learning approach for Sentiment Analysis of Hindi-English mixed language using an ensemble of CNN and LSTM networks. The proposed approach achieved an accuracy of 76.84% on the test set. The work in [5] addresses the prevalent use of code-mixed language questions in social media. This paper introduces an online framework for answering mixed language questions, decoding English mixed with Indian languages (Hindi, Telugu, Tamil). Employing deep learning techniques such as Recurrent Neural Network (RNN) and Hierarchical Attention Network (HAN), the system classifies questions.

The research in [6] tackles the problem of sentiment analysis on Hinglish YouTube comments using a novel Transformer-based model. The authors employed BERT, pre-trained on a vast text corpus and fine-tuned on the cooking dataset, achieving an 88.37% accuracy on the test set. The paper [7] proposes a new method for detecting hate speech in Hinglish code-mixed data. Character-level embeddings capture word context, these embeddings are then used to train a Gated Recurrent Unit (GRU) with an attention layer. The model attains a 92.5% accuracy on a test set comprising 10,000 labelled tweets and Facebook posts. Study [8] proposes a novel approach to sentiment analysis of mixed-case language Hinglish using natural language processing. The proposed system uses a two-stage approach: In the first stage, a lexicon-based approach is used to extract features from the tweets. In the second stage, SVM is used to classify the tweets into positive, negative, or neutral sentiment. It achieves an accuracy of 85% on the test dataset. [9] addresses sentiment analysis in code-mixed Kannada-English social media content. It introduces a self-created corpus from Facebook comments and utilizes SAIL's code-mixed datasets. The study evaluates distributed representation methods and compares various machine learning and deep learning techniques for sentiment analysis in this complex linguistic context.

Paper [10] proposes a deep learning approach for sentiment analysis and offensive language identification on multilingual code-mixed data. The proposed approach uses a pre-trained language model BERT to extract features from the data. These features are then used to train a classifier to predict the sentiment or offensiveness of the data. The dataset used in this study is the DravidianLangTech@ACL2022 dataset. This dataset contains 4076 code-mixed comments in Tamil and English. The comments are labelled with their sentiment (positive, negative, or neutral) and whether they contain offensive language. The proposed approach achieved an accuracy of 65% for sentiment analysis and 79% for offensive language identification. Another work [11] addresses the problem of the classification of Tamil movie tweets into positive, negative, or neutral sentiment. This is a challenging task due to the informal nature of tweets and the use of Tamil, which is a morphologically rich language. The authors collected a dataset of 7,842 Tamil movie tweets from Twitter. The tweets were manually labelled with positive, negative, or neutral sentiment. The authors evaluated the performance of their sentiment classifier on a test set of 1,568 tweets. The classifier achieved an accuracy of 83.5%.

A deep learning-based approach for text analytics of Malayalam-English code-mixed text was proposed in [12]. This approach uses a combination of word embedding, bidirectional LSTM, and attention mechanisms to learn the context of words in code-mixed text. It was evaluated on a dataset of Malayalam-English code-mixed text from social media. The results showed that the proposed approach achieved an F1-score of 0.76 for sentiment analysis and an F1-score of 0.99 for offensive language identification. Paper [13] introduces the problem of sentiment analysis from imbalanced code-mixed data and proposes a solution that uses a combination of sampling and machine learning techniques. The authors used a dataset of code-mixed tweets from India and balanced the dataset using SMOTE. The dataset was balanced, with 50% positive, 25% negative, and 25% neutral tweets. The authors then used various machine learning techniques to perform sentiment analysis and achieved an F1-score of 82.4%, which is a significant improvement over the baseline F1-score of 68.2%. Paper [14] addresses the challenges of developing NLP systems for low-resource languages like Telugu. The authors compiled a large Telugu corpus of over 7 million sentences and constructed sentiment, emotion, and hate speech lexicons. Multiple NLP models were trained, including DistilBert, on this corpus and achieved state-of-the-art results on sentiment analysis, emotion identification, hate speech, and sarcasm detection tasks. DistilBert achieved remarkable accuracy of 92.2% for sentiment analysis, 88.4% for emotion identification, 95.1% for hate speech detection, and 94.3% for sarcasm detection. The authors emphasize the importance of large corpora and efficient training methods for developing high-performance NLP systems for low-resource languages.

The study [15] proposes a method for using code-mixed data analysis to extract real-time traffic insights from Twitter. The authors used a dataset of 10 million geo-tagged UAE tweets and employed a two-step process: an NLP algorithm (Stanford CoreNLP) to identify traffic-related tweets and a Random Forest ML algorithm to categorize them (e.g., "traffic jam," "accident"). Google Maps API was used to visualize the traffic events. The method achieved 92% accuracy in jam detection and 85% in tweet categorization, demonstrating its potential as a valuable real-time tool for traffic analysis and management. The study [16] leverages Twitter data to predict traffic conditions in Indian cities like Bengaluru, Mumbai, and Delhi. By correlating tweet data, event information, and geographical locations, the research employs Natural Language Processing and machine learning techniques. The approach achieved 96% accuracy using DistilBertTokenizer + DistilBert classifier and provides insights into traffic congestion causes, enhancing real-time traffic information and disaster response. The limitation lies in its reliance on Twitter reporting for detecting unreported traffic jams.

The authors [17] investigated the possibility of enhancing pre-trained multilingual language models (PMLMs) for code-mixed text categorization. They proposed a new method that uses adapters to achieve state-of-the-art results on a Sinhala-English code-mixed dataset. 10,000 user comments were included in the dataset, which had five classification tasks carefully annotated: sentiment analysis, humor detection, hate speech detection, language identification, and aspect identification. The proposed method achieved an accuracy of 92.4% on the sentiment analysis task, 91.2% on the humor detection task, 90.8% on the hate speech detection task, 91.5% on the language identification task, and 89.6% on the aspect's identification task. Although it gives good result, their method requires a pre-trained PMLM, which may not be available for all languages, and its performance may degrade if the code-mixed data is noisy or unbalanced. Similarly, paper [18] offers a concise overview of code-mixing, discussing its forms, impacts, and relevance in linguistic studies and technology. Both papers contribute insights into sentiment analysis and code-mixing, providing valuable directions for future research.

The paper [19] proposes a transformer-based approach to language identification (LI) for Malayalam-English code-mixed text. The proposed approach uses a pre-trained transformer model, BERT, to learn the representation of words in both languages. The model is then fine-tuned on a "Malayalam-English Code-Mixed dataset". The authors evaluate their approach on a held-out test set and achieved an F1-score of 90.5%. The authors believe that the proposed approach is a promising method for LID for Malayalam-English code-mixed text. Future work could explore other transformer-based models for LID, explore other ways to fine-tune transformer models for LID, and develop a system that can automatically identify the language of code-mixed text. Further, study [20] proposes a machine learning approach for language identification (LID) in code-mixed "Hinglish and Urdu-English" social media text. Multinomial Naïve Bayes, Decision Tree, and Support Vector Machine algorithms were used on a dataset of 10,000 tweets collected from Twitter. SVM achieves the highest accuracy at 92.4%, showing promise for LID. However, rare words in code-mixed text can affect performance. Future work involves larger datasets, other languages, and real-world applications. The work in [21] addresses language identification in code-mixed social media content, crucial due to its prevalence on platforms like Facebook. Word embedding methods like CBOW and Skip-Gram generate feature vectors for machine learning algorithms (SVM, Logistic Regression, etc.), resulting in favorable cross-validation scores.

This work acknowledges the work already done and tries to carry forward the research done by addressing the limitation observed in previous works for code mixed translation for Indian languages Hindi-English, Malayalam-English, Tamil-English, Telugu-English by finetuning pretrained models like Indic LID for Language Identification (LID) module and use an ensemble approach for transliteration + translation using Indictrans and IndicXlit for code mixed machine which will be given as input to classification algorithm which performs Sentiment Analysis and predict the sentiment.

3. Dataset

Indic LID, developed by IIT Madras, has been meticulously trained on an extensive parallel text and code collection spanning 22 Indian languages. Termed the "Bhasha-Abhijnaanam" dataset [22][23], it's a creation of the "AI4Bharat" initiative at the Indian Institute of Technology Madras. Comprising a staggering 100 million words of text and code, this dataset is equitably distributed across all 22 languages. The foundation of Bhasha-Abhijnaanam dataset is rooted in diverse sources. It amalgamates publicly available datasets like IndicCorpus and Wikipedia dumps alongside internal datasets such as the NLLB dataset. Simultaneously, the IndicTrans [24] model is nurtured on a data repository encompassing text and code in 12 Indic languages. Dubbed "IndicTrans-2023", this dataset is a creation of the "AI4Bharat" project at IIT Madras. Boasting over 200 million words of text and code, IndicTrans-2023 exhibits a well-distributed representation across its 12 constituent languages.

These pretrained models lay the groundwork for a sophisticated ensemble approach in code-mixed translation. Transitioning to the sentiment analysis facet, a composite dataset of code-mixed tweet data was created from different online sources. This dataset, classified into three categories: positive, negative, and neutral, encompasses languages such as Hindi-English, Telugu-English, Tamil-English, and Malayalam-English. Derived from diverse sources and incorporating tweets and movie reviews, this dataset facilitates sentiment analysis experimentation and model refinement. Collectively, these resources and datasets form the basis of a comprehensive approach to language identification, transliteration, translation, and sentiment analysis for code-mixed content.

4. Proposed Approach

The entire process of sentiment analysis for code-mixed text involves a systematic series of steps. Initially, a dataset of tweets categorized into positive, negative, and neutral sentiments is collected and prepared through web scraping. Subsequently, the dataset undergoes pre-processing to ensure it's well-suited for training purposes. This involves steps like cleaning and removing unwanted elements like punctuation, HTML tags, and URLs, lowercasing, tokenization, removing duplicates, stop word removal, Stemming/Lemmatization etc. Following this, a Language Identification (LID) module comes into play, adept at identifying the language within the input code-mixed data, thereby providing language tags as outputs. This task is facilitated by employing the IndicLID model, specifically designed for this purpose. With language identification accomplished, the corresponding language of each sentence

is determined. Now, the IndicXlit module takes charge, utilizing the detected language to transliterate the code-mixed text back to its original form. This step is critical for maintaining accuracy and consistency. Once transliteration is complete, the resultant text is fed into the IndicTrans system, responsible for translating the content into monolingual English, ensuring uniformity across the dataset. At this stage, the groundwork has been laid to train a sentiment analysis model from scratch. Diverse algorithms, such as BERT-Multilingual, XLNet, and DistilBert, are employed for vector generation post tokenization. To fine-tune the model, an array of hyperparameters is experimented with, followed by the initialization and training process using classifiers like DistilBert, XLNet, and BERT. In the final stages, the trained model is subjected to testing and validation using sample sentences categorized into positive, negative, and neutral sentiments. This ensures the model's accuracy and functionality. Importantly, the performance and results of the developed model are compared and contrasted with those generated by the Google Machine Translate output. This comprehensive design and execution sequence encompass data collection, pre-processing, language identification, transliteration, translation, model training, hyperparameter tuning, testing, validation, and performance benchmarking, ultimately culminating in a refined and evaluated sentiment analysis solution tailored for code-mixed text. The proposed approach has been visualized in Fig. 1.

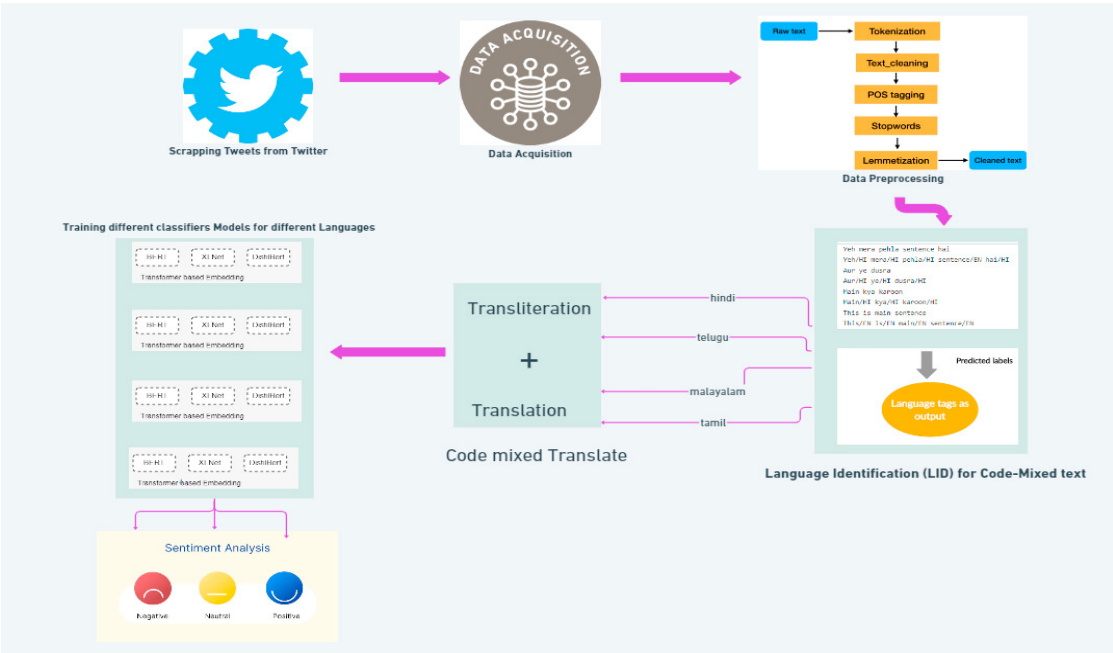

Fig. 1. Design of Proposed approach

5. Methodology

Detailed step-wise methodology for the proposed algorithm/approach for sentiment analysis of code-mixed text is as follows:

- Data Collection and Preparation**:** Collect a dataset of tweets categorized into positive, negative, and neutral sentiments through web scraping or other suitable means. Clean the dataset to remove any irrelevant or noisy data, such as special characters, URLs, and duplicate entries.
- Data Pre-processing**:** Tokenize the text by splitting it into words or sub word tokens. Apply text normalization techniques such as lowercasing, removing stop words, and stemming or lemmatization to standardize the text. Perform data augmentation if necessary, to balance the class distribution in the dataset.
- Language Identification (LID):
  - Implement a Language Identification (LID) module: Utilize a robust LID module like IndicLID [22] to identify the languages within the code-mixed text. IndicLID is a language identification model trained

on a massive dataset of over 100 million sentences, achieving an accuracy of over 98% for both native-script and Romanised text.
    - Assign language tags to each sentence or text segment: After identifying the languages, assign appropriate language tags to each sentence or text segment. These tags will serve as markers for subsequent processing steps.
- Transliteration:
    - Utilize the detected language tags: Leverage the language tags obtained from the LID step to transliterate the code-mixed text back to its original form.
    - Employ the IndicXlit model [23]: Utilize IndicXlit, a multilingual transformer-based model developed by the AI4Bharat community, to perform transliteration. IndicXlit supports 21 Indic languages and is trained on a vast dataset of 26 million-word pairs.
- Translation to English:
    - Translate the transliterated text into monolingual English: Employ the IndicTrans model [24], an advanced neural machine translation (NMT) model, to translate the transliterated text into English. IndicTrans facilitates translation between English and various Indic languages.
    - Ensure uniformity in the dataset: By translating the text into English, the dataset is unified, ensuring consistency for sentiment analysis.
- Model Selection and Vector Generation: Choose a sentiment analysis model such as BERT-Multilingual, XLNet, or DistilBert. Tokenize the translated text and use the chosen model to generate word embeddings or vectors for each token in the text.
- Model Fine-tuning and training: Experiment with an array of hyperparameters for the selected model, including learning rate, batch size, and the number of training epochs. Fine-tune the model using labelled data to adapt it to the specific task of sentiment analysis. Initialize and train the sentiment analysis model using the fine-tuned model architecture, taking into account the language tags for multi-language sentiment analysis. Testing and Validation: Test the trained model's performance on a set of sample sentences categorized into positive, negative, and neutral sentiments. Compare and contrast the performance and results of the developed model with those generated by the Google Machine Translate output. Evaluate the accuracy, precision, recall, F1-score, and other relevant metrics to ensure the model's functionality.

6. Results

First, the accuracy of the IndicLID was tested in terms of language Identification and it was able to identify Indic languages with an accuracy of over 99% but identify code mixed Hindi-English, Tamil- English, Telugu- English and Malayalam-English with an accuracy of 95%. These accuracies could be seen in Table 1.

Further compared the correctness of IndicLID + IndicTrans framework against Google Machine Translation in terms of the translating the code-mixed text in the four above mentioned languages with a benchmark test set created by us. This test set included 500 code mixed sentences along with their correct translations in English and observed that Code Mixed Translation Framework (ensemble model of IndicLID + IndicTrans) showed better accuracy in correct translation of code-mixed data as compared to resulting text translated by Google translate. This is because the Code-Mixed Translation Framework (ensemble model of IndicLID + IndicTrans) uses a Transliteration + translation approach whereas Google translate simply translates the text to make it understandable in the target language which might be challenging in case of Romanised code mixed text. Transliteration + translation involves transliterating the text into the target script, and then translating the text into the target language. This can be a more accurate and effective way to communicate with people who do not speak your language. It can help to preserve the original meaning of the text.

For Sentiment Analysis on Code Mixed Data, several combinations of machine learning and deep learning models were evaluated, as well as multiple word embedding models, in this study. The achieved accuracies are summarized in the table below. The observations in this study indicate that ML algorithms like RandomForestClassifier, LogisticRegression, and MultinomialNB with TfidfVectorizer produced average classification results, while SVM yielded the best outcomes. The reason for including these Machine learning algorithms was because it is two class classification and these are simple models to test with. Transformer models,

including BERT, also demonstrated comparable accuracy results, but the combination of DistilBertTokenizer + DistilBert classifier delivered the highest accuracy for both cases.

Table 1. Accuracies achieved on different classification algorithms for Sentiment Analysis on Code mixed data:

		Hi-En	Ta-En	Te-En	Ml-En	Mixed
RF	Normal	48 %	74 %	65 %	71 %	34 %
	GMT	57 %	88 %	80 %	89 %	58 %
	CMT	60%	88%	85%	89.5%	60%
SVC	Normal	63 %	71.5 %	76.5 %	82 %	69 %
	GMT	84 %	88 %	84 %	90 %	74 %
	CMT	90%	90%	89%	91%	80%
MNB	Normal	62 %	74.2 %	74 %	81 %	54 %
	GMT	75 %	80 %	77%	84 %	75 %
	CMT	75%	81%	79%	85%	79%
LR	Normal	66 %	75 %	77 %	79 %	58 %
	GMT	73 %	78 %	80 %	82 %	72 %
	CMT	70%	77%	81%	83%	75%
XL-NET	Normal	83 %	87 %	78 %	88 %	72 %
	GMT	85 %	88 %	82 %	89 %	79 %
	CMT	89 %	90%	85%	89%	80%
Bert-base-Multilingual	Normal	85 %	89 %	86.5 %	87 %	79 %
	GMT	90 %	92%	89%	90%	84%
	CMT	91.5 %	91.5%	90.5%	89%	89%
DistilBERT	Normal	85 %	89%	86 %	87.5 %	80%
	GMT	91 %	92%	90%	90%	86.5%
	CMT	92 %	94%	90%	92%	90%

Google Machine Translation (GMT) data is the data on which Google translate has been used before giving as an input to Classification algorithms.

Code Mixed Translation (CMT) data is the data on which IndicLID + IndicTrans+ IndicXLit model is applied and Normal data is the raw data on which no translation has been applied before Classification.

It is important to note that these accuracy results are based on specific datasets that was used in this work and is approximated to average value observed during several runs and may not be representative of IndicLID, IndicTrans and IndicXlit's performance on other datasets. Additionally, the accuracy of these models may also vary depending on the length and complexity of the text snippet being translated using Google Translate or the ensemble mode.

7. Conclusion and Future Work

The study presented a novel approach for translating code-mixed data in Indian languages, including Hindi, Malayalam, Tamil, and Telugu, which can be leveraged as input for NLP applications such as Sentiment Analysis. The approach utilized fine-tuned pretrained models like IndicLID for Language Identification (LID) and employed an ensemble approach for transliteration and translation using IndicTrans and IndicXlit. This translated data was then input to a classification algorithm for Sentiment Analysis. The study also compared this approach with the popular machine translation tool, Google Translate, and observed that the proposed transformer model performed comparably or even better than traditional machine translation for the specified Indian languages. The framework employed in this work offers a set of notable advantages when it comes to transliterating and identifying Indic languages in code-mixed data. First and foremost, it excels in accuracy, demonstrating exceptional precision in its

ability to transliterate and identify a diverse array of Indian languages, including but not limited to Hindi, Gujarati, Tamil, Telugu, Malayalam, and many more. This high level of accuracy is paramount in maintaining the authenticity and meaning of code-mixed text, ensuring reliable and consistent results. Furthermore, this framework stands out due to its multilingual capabilities. Indic Trans + IndicLID serves as a versatile and adaptable system that can handle the transliteration and identification of multiple Indic languages. This framework is well-suited for code-mixed data, where different languages are seamlessly integrated into a single context. Given the prevalence of code-mixing in modern communication, particularly in social media, the framework's capacity to handle code-mixed text is highly advantageous, ensuring its relevance and effectiveness in real-world language scenarios.

However, it was noted that the proposed approach has its limitations, primarily regarding computational efficiency. The ensemble approach, especially when processing large amounts of text, proved to be slow. Additionally, the study was limited to having one of the languages in the code-mixed data as English to produce output in English, while the others could be any of the four specified Indian languages. Furthermore, it was noted that the approach assumes only one code-mixed language per sentence. This could be a limitation in real-world scenarios where multiple languages may be used within a single sentence. Finally, the study acknowledged that the system may provide inaccurate results if the spellings of Romanized code-mixed input are incorrect or not commonly used. In conclusion, this study lays the framework for leveraging NLP techniques to address the challenges and opportunities presented by code-mixed text in Indian Languages.

Future work would include investigating methods to enhance the computational efficiency of the ensemble approach, including the possibility of training a more streamlined model that integrates Language Identification, transliteration, and translation, extend the system's capabilities to handle sentences with multiple code-mixed languages more effectively, catering to the complexity of real-world communication, exploring the inclusion of additional Indian languages and combinations, addressing the linguistic diversity of the Indian subcontinent and potentially expanding to other multilingual regions, integrating mechanisms for spell-checking and error correction to improve the system's robustness in handling variations and non-standard spellings in code-mixed text and extending the utilization of code-mixed translation to other NLP applications, such as chatbots and virtual assistants, further validating the framework's utility.

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